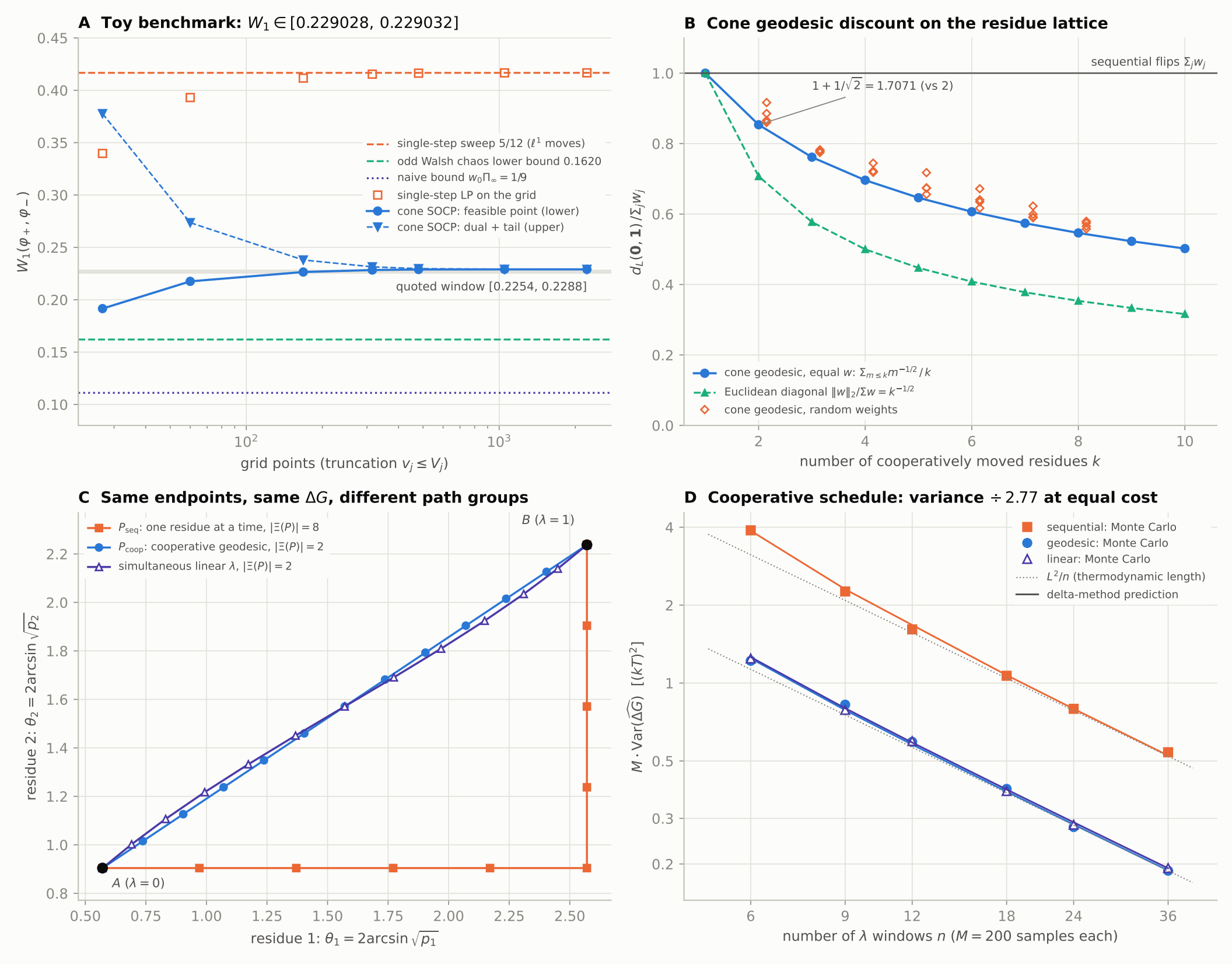

A Toy benchmark: $W_1 \in [0.229028, 0.229032]$
single-step sweep 5/12 (ℓ^1 moves)
odd Walsh chaos lower bound 0.1620
naive bound $w_0\Pi_\infty = 1/9$
single-step LP on the grid
cone SOCP: feasible point (lower)
cone SOCP: dual + tail (upper)
quoted window [0.2254, 0.2288]
$W_1(\varphi_+, \varphi_-)$
grid points (truncation $v_j \le V_j$)
B Cone geodesic discount on the residue lattice
sequential flips $\Sigma_j w_j$
$1 + 1/\sqrt{2} = 1.7071$ (vs 2)
cone geodesic, equal w: $\Sigma_{m \le k} m^{-1/2}/k$
Euclidean diagonal $\|w\|_2/\Sigma w = k^{-1/2}$
cone geodesic, random weights
$d_L(\mathbf{0}, \mathbf{1})/\Sigma_j w_j$
number of cooperatively moved residues k
C Same endpoints, same ΔG, different path groups
P_{seq}: one residue at a time, $|\Xi(P)| = 8$
P_{coop}: cooperative geodesic, $|\Xi(P)| = 2$
simultaneous linear λ, $|\Xi(P)| = 2$
B ($\lambda = 1$)
A ($\lambda = 0$)
residue 2: $\theta_2 = 2\arcsin\sqrt{p_2}$
residue 1: $\theta_1 = 2\arcsin\sqrt{p_1}$
D Cooperative schedule: variance ÷ 2.77 at equal cost
sequential: Monte Carlo
geodesic: Monte Carlo
linear: Monte Carlo
L^2/n (thermodynamic length)
delta-method prediction
$M \cdot \mathrm{Var}(\widehat{\Delta G})$ $[(kT)^2]$
number of λ windows n ($M = 200$ samples each)